Model Question Paper-I with effect from 2023-24 (CBCS Scheme)

USN | | | | | | | | | | |

Third Semester B.E. Degree Examination
Mathematics for Computer Science

TIME: 03 Hours **Max. Marks: 100**

Note: 01. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**.
02. Statistical tables and Mathematics formulae handbooks are allowed

		Module -1	Bloom's Taxonomy Level	Marks
Q.01	a	The probability distribution function of variate X is given by the following table; x: 0 | 1 | 2 | 3 | 4 | 5 | 6 P(x): k | 3k | 5k | 7k | 9k | 11k | 13k i) Find the value of k, ii) $P(x \geq 5)$ & iii) $P(3 < x \leq 6)$.	L2	6
	b	If the probability of a bad reaction from a certain injection is 0.001, determine the chance that more than two of 2000 individuals will have a bad reaction.	L3	7
	c	Find the mean and standard deviation of Poisson's distribution.	L2	7
		OR		
Q.02	a	The probability of a pen manufactured by a factory be defective is 1/10. If 12 such pens are manufactured, what is the probability that i) exactly 2 are defective, ii) at least 2 are defective, iii) none of them are defective.	L3	6
	b	Determine the value of k, so that the function $f(x) = k(x^2 + 4)$ for x = 0, 1, 2, 3 can serve as a probability distribution of the discrete random variable X: Also find i) $P(0 < x \leq 2)$ and ii) $P(x \geq 1)$.	L2	7
	c	Find the mean and standard deviation of Binomial distribution.	L2	7
		Module-2		
Q.03	a	The joint probability distribution of discrete random variables X & Y are as follows; X\Y: -3 | 2 | 4 1: 0.1 | 0.2 | 0.2 2: 0.3 | 0.1 | 0.1 Then i) determine marginal distribution of X & Y, ii) show that X & Y are dependent.	L2	6
	b	Determine the value of k so that the function $f(x, y) = k\|x - y\|$, for $x = -2, 0, 2$; $y = -2, 3$ represents joint probability distribution of the random variables X and Y. Also determine cov(X, Y).	L2	7
	c	Three boys X, Y, Z are throwing a ball to each other. X always throws the ball to Y & Y always throws the ball to Z. But Z is just as likely to throw the ball to Y or as to X. Write TPM if Z is the first person to throw the ball, find the probability that X has the ball after fourth throw.	L3	7
		OR		

<table>
<tr><td>Q.04</td><td>a</td><td>Given the following joint distribution of the random variable X & Y,
<table>
<tr><td>X\Y</td><td>−2</td><td>−1</td><td>4</td><td>5</td></tr>
<tr><td>1</td><td>0.1</td><td>0.2</td><td>0</td><td>0.3</td></tr>
<tr><td>2</td><td>0.2</td><td>0.1</td><td>0.1</td><td>0</td></tr>
</table>
Determine the marginal probability distributions of X & Y. Also compute i) Expectations of X, Y & XY, ii) Covariance of X & Y, iii) Correlation of X & Y.</td><td>L2</td><td>6</td></tr>
<tr><td></td><td>b</td><td>The joint probability distribution of random variables X & Y are as follows.
<table>
<tr><td>x\y</td><td>−4</td><td>2</td><td>7</td></tr>
<tr><td>1</td><td>1/8</td><td>¼</td><td>1/8</td></tr>
<tr><td>5</td><td>¼</td><td>1/8</td><td>1/8</td></tr>
</table>
then determine i) marginal distribution of X & Y, ii) E(X), E(Y) & E(XY), iii) COV(X, Y), iv) ρ(X, Y).</td><td>L2</td><td>7</td></tr>
<tr><td></td><td>c</td><td>The students study habits are as follows; If he studies on one night, he is 60% sure not to study on next night. On the other hand, if he does not study on to night, he is 80% sure to study next night. Write the transition probability matrix for his chain of study. In the long run how often does he study? Suppose he studies on Monday night, what is the probability that he does not study on Friday night?</td><td>L3</td><td>7</td></tr>
<tr><td colspan="4">Module-3</td></tr>
<tr><td>Q.05</td><td>a</td><td>A die was thrown 9000 times and throw of 5 or 6 was obtained 3240 times on the assumption of random throwing do the data indicate an unbiased die?</td><td>L3</td><td>6</td></tr>
<tr><td></td><td>b</td><td>Before an increase in excise duty on tea 400 people out of a sample 500 persons were found to be tea drinkers. After an increase in duty 400 people were tea drinkers in a sample of 600 people. Using standard error of proportion, state whether there is a significant decrease in the consumption of tea for 95% and 99% level of significance?</td><td>L3</td><td>7</td></tr>
<tr><td></td><td>c</td><td>A survey was conducted in a slum locality of 2000 families by selecting a sample of size 800 families. It was revealed that 180 families were illiterate. Find the probable limits of the illiterate families in the population of 2000.</td><td>L3</td><td>7</td></tr>
<tr><td colspan="4">OR</td></tr>
<tr><td>Q.06</td><td>a</td><td>In 324 throws a die an odd number turned up 181 times. Is it reasonable to think that the die is an unbiased one?</td><td>L3</td><td>6</td></tr>
<tr><td></td><td>b</td><td>The mean weight obtained from a random sample of size 100 is 64gms. The standard deviation of the weight distribution of the population is 3gms. Test the statement that the mean weight of the population is 67gms at 5% level of significance. Also set up 99% confidence limits of the mean weight of the population.</td><td>L3</td><td>7</td></tr>
<tr><td></td><td>c</td><td>In a sample of 100 people in this city, the average income was Rs. 210, with a standard deviation of Rs. 10. For another sample of 150 persons, the average income was Rs. 220, with a standard deviation of Rs. 12. The standard deviation of the incomes of the people of the city was Rs. 11. Test whether there is any significant difference between the average incomes of the localities.</td><td>L3</td><td>7</td></tr>
</table>

Module-4

Q.07 a. An experiment on Pea breading the following frequency of seeds were obtained — **L3** — **6**

Round & Yellow	Wrinkled & Yellow	Round & Green	Wrinkled & Green	Total
315	101	108	32	556

Theory predicts that the frequences should be in the proportions 9:3:3:1. Examine the correspondence between theory and experiment ($\chi^2_{0.05}$= 7.815).

b. Use the Central Limit theorem to evaluate $P[50 < \bar{X} < 56]$ where $\bar{X}$ represents the mean of a random sample of size 100 from an infinite population with mean $\mu = 53$ and variance $\sigma^2 = 400$ (Given, A(1.5) = 0.4332). — **L2** — **7**

c. Ten individuals are chosen at random from a population and their heights in inches found to be 63, 63, 66, 67, 68, 69, 70, 70, 71 and 71. Test the hypothesis that the mean height of the universe is 66 inches. ($t_{0.05}$ = 2.262 for 9 d.f.). — **L3** — **7**

OR

Q.08 a. The following table shows the runs scored by two batsmen can it be said that the performance of batsman A is more consistent than the performance of batsman B? Use 1% level of significance ($F_{0.01}$,4,7 = 7.85). — **L3** — **6**

Batsman-A	40	50	35	25	60	70	65	55
Batsman-B	60	70	40	30	50			

b. The following table gives the number of aircraft accidents that occurred during the various days of the week. Find whether the accidents are uniformly distributed over the week? — **L3** — **7**

Days	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Total
Accidents	14	16	8	12	11	9	14	84

Given that $\chi^2_{0.05}$= 12.59.

c. Consider the sample consisting of nine numbers 45, 47, 50, 52, 48, 47, 49, 53, 51. The sample is drawn from a population whose mean is 47.5. Find whether the sample mean differs significantly from the population mean at 5% level of significance ($t_{0.05}$ for 8 d.f. = 2.31). — **L2** — **7**

Module-5

Q.09 a. A manufacturing company has purchase three new machines of different brands and wishes to determine whether one of them is faster than the others in producing a certain output, 5 hourly production figures are obtained at random from each other machine and the results are given below; — **L3** — **10**

Observation	A	B	C
1	25	31	24
2	30	39	30
3	36	38	28
4	38	42	25
5	31	35	28

Use ANOVA and determine whether the machines are significantly

different in their mean speed. Given at 5% level $F_{2,12} = 3.89$.

b. Set up ana analysis of variance table for the following per acre production data for three varieties of wheat, each grown on 4 plots and state if the variety differences are significant.

Plot of Land	Per acre production data		
	A	B	C
1	6	5	5
2	7	5	4
3	3	3	3
4	8	7	4

Use ANOVA, given at 5% level $F_{2,9} = 4.26$.

L3 | 10

OR

Q.10 a. Set up an analysis of variance table for the following two-way design results: per acre production data of wheat in metric tons;

Varieties of fertilizers	Varieties of seeds		
	A	B	C
W	6	5	5
X	7	5	4
Y	3	3	3
Z	8	7	4

Also state whether variety differences are significant at 5% level. Given that $F_{2,6} = 5.14$ and $F_{3,6} = 4.76$.

L3 | 10

b. Analyze the variance in the following table Latin square of yields in kgs of Paddy where A, B, C, D denotes the different methods of cultivation.

D-122	A-121	C-123	B-122
B-124	C-123	A-122	D-125
A-120	B-119	D-120	C-121
C-122	D-123	B-121	A-122

Examine whether the different methods of cultivation have given significantly different is given that $F_3 = 4.76$.

L3 | 10

Model Question Paper-I (CBCS Scheme)

USN										

Third Semester B.E Degree Examination

MATHEMATICS FOR CS ENGINEERING STREAM (BCS301)

TIME: 03 Hours **Max.Marks:100**

Note: (i) Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE.**
(ii) Statistical tables and Mathematics Formula handbooks are allowed.

		Module -1	M	L	C
Q.01	a	A shipment of 8 similar microcomputers to a retail outlet contains 3 that are defective. If a school makes a random purchase of 2 of these computers, find the probability distribution for the number of defectives. Find the mean and variance of this distribution.	**6**	**L2**	**CO1**
	b	In a factory producing blades, the probability of any blade being defective is 0.002. If blades are supplied in packets of 10, Using Poisson distribution determine the number of packets containing i. No defective ii. One defective and iii. Two defective blades respectively in a consignment of 10000 packets	**7**	**L2**	**CO2**
	c	If the mileage (in thousands of miles) of a certain radial tyre is a random variable with exponential distribution with mean 40,000 miles. Determine the probability that the tyre will last i. At least 20,000 miles ii. At most 30,000 miles	**7**	**L3**	**CO2**
		OR			
Q.02	a	The density function of a random variable X is given by $f(x) = \begin{cases} k\sqrt{x}, & 0 < x < 1 \\ 0, & elsewhere \end{cases}$, i. Find k ii. Find the cdf F(x) and use it to evaluate $P[0.3 < X < 0.6]$	**6**	**L2**	**CO1**
	b	Find the mean and variance of Binomial distribution.	**7**	**L2**	**CO1**

	c	In a test on 2000 electric bulbs, it was found that the life of a particular make , was normally distributed with an average life 2040 hours and standard deviation of 60 hours. Estimate the number of bulbs likely to burn for i. More than 2150 hours ii. Less than 1950 hours iii. In between 1920 and 2160 hours	7	L3	CO2
		Module-2			
Q. 03	a	The joint probability distribution of two random variables X and Y is <table><tr><td>Y \ X</td><td>-4</td><td>2</td><td>7</td></tr><tr><td>1</td><td>1/8</td><td>1/4</td><td>1/8</td></tr><tr><td>5</td><td>1/4</td><td>1/8</td><td>1/8</td></tr></table> Find the marginal distributions. Obtain the correlation coefficient between X and Y.	6	L2	CO1
	b	Find the unique fixed probability vector of $P = \begin{pmatrix} 0 & \frac{3}{4} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & 0 \end{pmatrix}$	7	L2	CO3
	c	Every year, a man trades his car for a new car. If he has a Maruti, he trades it for an Ambassador. If he has an Ambassador, he trades it for Santro. However, if he had a Santro, he is just as likely to trade it for a new Santro as to trade it for a Maruti or an Ambassador. In 2000 he bought his first car which was a Santro. Find the probability that he has i. 2002 Santro ii. 2002 Maruti iii. 2003 Ambassador iv. 2003 Santro	7	L3	CO3
		OR			
Q.04	a	The joint probability distribution of two random variables X and Y is <table><tr><td>Y \ X</td><td>-3</td><td>2</td><td>4</td></tr><tr><td>1</td><td>0.1</td><td>0.2</td><td>0.2</td></tr><tr><td>3</td><td>0.3</td><td>0.1</td><td>0.1</td></tr></table> i. Are X and Y independent?	6	L2	CO1

		ii. Evaluate $P[Y \leq 2]$ iii. Evaluate $P[X+Y \leq 2]$			
	b	Define Probability Vectors, Stochastic matrices, Regular stochastic matrix, stationary distribution and absorbing state of Markov chain.	7	L1	CO3
	c	A salesman's territory consists of 3 cities A, B and C. He never sells in the same city on successive days. If he sells in city A, then the next day he sells in city B. However, if he sells in either B or C, then the next day he is twice as likely to sell in city A as in other city. In long run, how often does he sell in each of the cities.	7	L3	CO3
		Module-3			
Q. 05	a	Define Null hypothesis, significance level, critical region, Type-I and type-II errors in a statistical test.	6	L1	CO4
	b	A coin was tossed 400 times and head turned up 216 times. Test the hypothesis that the coin is un biased at 5% level of significance.	7	L3	CO4
	c	In a city A 20 % of a random sample of 900 school boys had a certain slight physical defect. In another city B, 18.5% of a random sample of 1600 school boys had the same defect. Is the difference between the proportions significant at 5% significance level	7	L3	CO4
		OR			
Q. 06	a	Explain the following terms i. Standard error ii. Statistical hypothesis iii. Critical region of a statistical test iv. Test of significance	6	L1	CO4
	b	A die was thrown 9000 times and a throw of 5 or 6 was obtained 3240 times. On the assumption of random throwing, do the data indicate an unbiased die at 1% level of significance.?	7	L3	CO4
	c	In a sample of 600 men from a certain city, 450 are found smokers. In another sample of 900 men from another city, 450 are smokers. Do the indicate that the cities are significantly different with respect to the habit of smoking among men. Test at 5% significance level.	7	L3	CO4
		Module-4			
Q. 07	a	State Central limit theorem. Use the theorem to evaluate $P[50 < \bar{X} < 56]$ where $\bar{X}$ represents the mean of a random sample of size 100 from an infinite population with mean $\mu = 53$ and variance $\sigma^2 = 400$	6	L2	CO5
	b	A random sample of size 25 from a normal distribution $N(\mu, \sigma^2 = 4)$ yields, sample mean $\bar{X} = 78.3$. Obtain a 99% confidence interval for μ.	7	L2	CO5
	c	A survey of 320 families with 5 children each revealed the following distribution.	7	L3	CO4

No. of boys	5	4	3	2	1	0

		No. of girls	0	1	2	3	4	5			
		No. of families	14	56	110	88	40	12			
		Is the result consistent with the hypothesis that male and female births are equally probable at 5% level of significance.									
		OR									
Q. 08	a	A random sample of size 64 is taken from an infinite population having mean 112 and variance 144. Using central limit theorem, find the probability of getting the sample mean $\bar{X}$ greater than 114.5							6	L2	CO5
	b	Let the observed value of the mean $\bar{X}$ of a random sample of size 20 from a normal distribution with *mean* μ and variance $\sigma^2 = 80$ be 81.2. Find a 90% and a 95% confidence intervals for μ.							7	L2	CO5
	c	The nine items of a sample have the following values: 45, 47, 50, 52, 48, 47, 49, 53, 51. Does the mean of these differ significantly from the assumed mean of 47.5 at 5% significance level?							7	L3	CO4
		Module-5									
Q. 09	a	Three different kinds of food are tested on three groups of rats for 5 weeks. The objective is to check the difference in mean weight (in grams) of the rats per week. Apply one-way ANOVA using a 0.05 significance level to the following data:							10	L3	CO6
		Food 1	**8**	**12**	**19**	**8**	**6**	**11**			
		Food 2	**4**	**5**	**4**	**6**	**9**	**7**			
		Food 3	**11**	**8**	**7**	**13**	**7**	**9**			
	b	Analyze and interpret the following statistics concerning output of wheat per field obtained as a result of experiment conducted to test four varieties of wheat viz. A, B, C and D under a Latin- square design							10	L3	CO6
		C 25	B 23	A 20	D 20						
		A 19	D 19	C 21	B 18						
		B 19	A 14	D 17	C 20						
		D 17	C 20	B 21	A 15						

		OR			
Q. 10	a	Set up an analysis of variance table for the following per acre production data for three varieties of wheat, each grown on 4 plots and state it the variety differences are significant at 5% significant level Per acre production data Plot of land / Variety of Wheat: A, B, C 1: 6, 5, 5 2: 7, 5, 4 3: 3, 3, 3 4: 8, 7, 4	**10**	**L3**	**CO6**
	b	Set up ANOVA table for the following information relating to three drugs testing to judge the effectiveness in reducing blood pressure for three different groups of people Group of people / Drug: X, Y, Z A: 14, 10, 11; 15, 9, 11 B: 12, 7, 10; 11, 8, 11 C: 10, 11, 8; 11, 11, 7 Do the drugs act differently? Are the different groups of people affected differently? Is the interaction term significant? Answer the above questions taking a significant level of 5%	**10**	**L3**	**CO6**

Model Question Paper-II (CBCS Scheme)

USN | | | | | | | | | | |

Third Semester B.E Degree Examination

MATHEMATICS FOR CS ENGINEERING STREAM (BCS301)

TIME: 03 Hours **Max.Marks:100**

Note: (i) Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**
(ii) Statistical tables and Mathematics Formula handbooks are allowed.

		Module -1	M	L	C
Q.01	a	A random variable X has the following probability function for various values of x x: 0, 1, 2, 3, 4, 5, 6, 7 $P(x)$: 0, K, $2k$, $2k$, $3k$, k^2, $2k^2$, $7k^2+k$ (i) Find the value of k (ii) Evaluate $P[X<6]$, $P[0<X<5]$, and $P[X\geq 6]$	6	L2	CO1
	b	Find the mean and variance of Poisson distribution.	7	L2	CO1
	c	In a certain town the duration of a shower is exponentially distributed with mean 5 minutes. What is the probability that a shower will last for? (i) 10 minutes or more (ii) less than 10 minutes (iii) between 10 and 12 minutes.	7	L3	CO2
		OR			
Q.02	a	Determine the value k, so that the function $f(x)=k(x^2+4)$, for $x=0,1,2,3$ can serve as a probability distribution of the discrete random variable X: Also, find (i) $P[0<X\leq 2]$ & (ii) $P[X\geq 1]$	6	L2	CO1
	b	Out of 800 families with 5 children each, how many would you expect to have (i) 3 *boys* (ii) At least one boy (iii) At most two boys, assuming equal probabilities for boys and girls.	7	L3	CO2
	c	In a normal distribution, 31% of the items are under 45 and 8% are over 64. Find the mean and Variance of the distribution.	7	L3	CO1
		Module-2			
Q. 03	a	If the joint probability distribution of X and Y is given by $f(x,y)=\frac{x+y}{30}$, for $x=0,1,2,3; y=0,1,2$ Find (i) $P[X\leq 2, Y=1]$, (ii) $P[X>2, Y\leq 1]$ (iii) $P[X>Y]$	6	L2	CO1

	b	Find the unique fixed probability vector for the regular stochastic matrix $$A = \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{6} & \frac{1}{2} & \frac{1}{3} \\ 0 & \frac{2}{3} & \frac{1}{3} \end{bmatrix}$$	7	L2	CO3
	c	A gambler's luck follows' a pattern. If he wins a game the Probability of winning the next game is 0.6. However, if he loses a game, the probability of losing the next game is 0.7. There is an even chance of the gambler winning the first game. (i) What is the probability of he winning the second game? (ii) What is the probability of he winning the third game? (iii) In the long run, how often he will win?	7	L3	CO3
		OR			
Q.04	a	Determine the value of k so that the function $f(x,y) = k\|x-y\|, for\ x = -2, 0, 2; y = -2, 3$ represents joint probability distribution of the random variables X and Y. Also determine $Cov(X,Y)$.	6	L2	CO1
	b	Show that the matrix $\begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \end{bmatrix}$ is a regular stochastic matrix	7	L2	CO3
	c	Three boys A, B and C are throwing a ball to each other. A is just as likely to throw the ball to B as to C. B always throws the ball to A, and C is just as likely to throw the ball to A as to B. Find the probability that C has the ball after three throws if now A has the ball.	7	L3	CO3
		Module-3			
Q. 05	a	Explain the following terms (i) Standard error (ii) Statistical hypothesis (iii) Critical region of a statistical test (iv) Test of significance	6	L1	CO4
	b	In 324 throws of a six faced die, an odd number turned up 181 times. Is it reasonable to think that the die is unbiased one at 5% level of significance?	7	L3	CO4
	c	In an examination given to students at a large number of different schools the mean grade was 74.5 and S.D grade was 8. At one particular school where 200 students took the examination the mean grade was 75.9. Discuss the significance of this result at both 5% and 1% level of significance.	7	L3	CO4
		OR			
Q. 06	a	Define (i) Alternative hypothesis (ii) A statistic (iii) Level of significance and (iv) Two-tailed test	6	L1	CO4

	b	A coin is tossed 1000 times and head turns up 540 times. Decide on the hypothesis that the coin is unbiased at 1% level of significance.	7	L3	CO4
	c	One type of air craft is found to develop engine trouble in 5 flights out of a total of 100 and another type in 7 flights out of a total of 200 flights. Is there a significance difference in the two types of air craft's so far as engine defects are concerned? Test at 5% significance level.	7	L3	CO4
		Module-4			
Q. 07	a	An unknown distribution has a mean of 90 and a standard deviation of 15. Samples of size $n = 25$ are drawn randomly from the population. Find the probability that the sample mean is between 85 and 92.	6	L2	CO5
	b	The heights of a random sample of 50 college students showed a mean of 174.5 centimeters and a standard deviation of 6.9 centimeters. Construct a 99% confidence interval for the mean height of all college students.	7	L2	CO5
	c	A die was thrown 60 times and the following frequency distribution was observed: Faces: 1 | 2 | 3 | 4 | 5 | 6 frequency: 15 | 6 | 4 | 7 | 11 | 17 Test whether the die is unbiased at 5% significance level.	7	L3	CO4
		OR			
Q. 08	a	In a recent study reported on the Flurry Blog, the mean age of tablet users is 34 years. Suppose the standard deviation is 15 years. Take a sample of size $n = 100$. Using central limit theorem, find the probability that the sample mean age is more than 30 years.	6	L2	CO5
	b	Suppose that 10, 12, 16, 19 is a sample taken from a normal population with variance 6.25. Find a 95 percent confidence interval for the population mean.	7	L2	CO5
	c	A random sample of 10 boys had the following I.Q.: 70, 120, 110, 101, 88, 83, 95, 98, 107, 100. Do these data support the assumption of a population mean I.Q. of 100 (at 5% level of significance)?	7	L3	CO4
		Module-5			
Q. 09	a	Three types of fertilizers are used on three groups of plants for 5 weeks. We want to check if there is a difference in the mean growth of each group. Using the data given below apply a one-way ANOVA test at 0.05 significant level Fertilizer-1: 6 | 8 | 4 | 5 | 3 | 4 Fertilizer-2: 8 | 12 | 9 | 11 | 6 | 8 Fertilizer-3: 13 | 9 | 11 | 8 | 7 | 12	10	L3	CO6

	b	Present your conclusions after doing analysis of variance to the following results of the Latin-square design experiment conducted in respect of five fertilizers which were used on plots of different fertility A 16 B 10 C 11 D 9 E 9 E 10 C 9 A 14 B 12 D 11 B 15 D 8 E 8 C 10 A 18 D 12 E 6 B 13 A 13 C 12 C 13 A 11 D 10 E 7 B 14	10	L3	CO6
		OR			
Q. 10	a	A trial was run to check the effects of different diets. Positive numbers indicate weight loss and negative numbers indicate weight gain. Check if there is an average difference in the weight of people following different diets using an ANOVA Table. Low Fat: 8, 9, 6, 7, 3 Low Calorie: 2, 4, 3, 5, 1 Low protein: 3, 5, 4, 2, 3 Low carbohydrate: 2, 2, -1, 0, 3	10	L3	CO6
	b	The following data show the number of worms quarantined from the GI areas of four groups of muskrats in a carbon tetrachloride anthelmintic study. Conduct a two-way ANOVA test. I: 33, 32, 26, 14, 30 II: 41, 38, 40, 23, 21 III: 12, 35, 46, 22, 11 IV: 38, 43, 25, 13, 26	10	L3	CO6

CBCS SCHEME

USN | | | | | | | | | | |

BCS301

Third Semester B.E./B.Tech. Degree Examination, June/July 2024
Mathematics for Computer Science

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. VTU Formula Hand Book is permitted.
3. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Obtain the mean and variance of Poisson distribution.	06	L2	CO2
	b.	Out of 800 families with 4 children each, how many families would be expected to have (i) 2 boys and 2 girls (ii) atleast one boy (iii) at most 2 girls. Assume equal probabilities for boys and girls.	07	L3	CO2
	c.	The length of telephone conversation in a booth has been an exponential distribution and found on an average to be 5 minutes. Find the probability that a random call made from this booth (i) ends less than 5 minutes (ii) between 5 and 10 minutes.	07	L2	CO2
		OR			
Q.2	a.	The probability distribution of a finite random variable X is given by X: -2, -1, 0, 1, 2, 3 P(X): 0.1, k, 0.2, 2k, 0.3, k (i) Find the value of k (ii) Variance (iii) $P(x \leq 2)$	06	L2	CO1
	b.	The number of accidents in a year to taxi drivers in a city follows a Poisson distribution with mean 3. Out of 1000 taxi drivers find approximately number of drivers with (i) more than 3 accidents in a year (ii) at most 2 accidents in a year.	07	L3	CO2
	c.	The marks of 1000 students in an exam follows normal distribution with mean 70 and standard deviation 5. Find the students whose marks will be (i) less than 65 (ii) between 65 and 75. A(1) = 0.3413.	07	L3	CO2
		Module – 2			
Q.3	a.	Given the following joint distribution of the random variables X and Y. Find the corresponding marginal distribution. Also compute the covariance. X \ Y: 1, 3, 9 2: 1/8, 1/24, 1/12 4: 1/4, 1/4, 0 6: 1/8, 1/24, 1/12	06	L3	CO2
	b.	A salesmen's territory consists of 3 cities A, B and C. He never sells in the same city for 2 consecutive days. If he sells in city A, then the next day he sells in city B. However if he sells in either B or C then the next day he is twice as likely to sell in city A as in the other city. In the long run how often does he sell in each of the cities.	07	L3	CO3
	c.	Show that $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{bmatrix}$ is a regular stochastic matrix. Also find the associated unique fixed probability vector.	07	L2	CO2

		OR			
Q.4	**a.**	Define probability vector, regular stochastic matrix, fixed prob vector.	**06**	**L1**	**CO3**
	b.	The joint probability distribution of two discrete random variables X and Y is $f(x, y) = k(2x + y)$, where x and y are integers such that $0 \le x \le 2$, $0 \le y \le 3$. i) Find the value of the constant k. ii) Show that the random variables X and Y are dependent iii) Find $P(X \ge 1, Y \le 2)$.	**07**	**L3**	**CO2**
	c.	A fair coin is tossed thrice. The random variables X and Y are defined as X = 0 or 1 according as head or tail occurs on the first toss, y-number of heads. Compute e(X, Y)	**07**	**L3**	**CO2**
		Module – 3			
Q.5	**a.**	Define statistical hypothesis, null hypothesis, Type-I error and Type-II error.	**06**	**L1**	**CO4**
	b.	In 324 throws of a six faced die an even number turned up 181 times. Is it reasonable to think that the 'die' is an unbiased one at 99% level?	**07**	**L2**	**CO4**
	c.	Before an increase in excise duty on tea, 800 people out of sample of 1000 were consumers of tea. After the increase in duty, 800 people were consumers of tea in a sample of 1200 persons. Find whether there is significant decrease in the consumption of tea after the increase in duty at 1%. (One tailed test at 1% is 2.33).	**07**	**L3**	**CO4**
		OR			
Q.6	**a.**	A coin is tossed 1000 times and head turns up 540 times. Decide on the hypothesis that the coin is unbiased.	**06**	**L2**	**CO4**
	b.	In an exit poll enquiry it was revealed that 600 voters in one locality and 400 voters from another locality favoured 55% and 48% respectively a particular party to come to power. Test the hypothesis that there is a difference in the locality in respect of the opinion.	**07**	**L3**	**CO4**
	c.	A random sample for 1000 workers in company has mean wage of Rs.50 per day and standard deviation of Rs.15. Another sample of 1500 workers from another company has mean wage of Rs.45 per day and standard deviation of Rs.20. Does the mean rate of wages varies between the two companies at 95% confidence limit?	**07**	**L3**	**CO4**
		Module – 4			
Q.7	**a.**	The mean life time of a sample of 25 bulbs is found as 1550 hrs with standard deviation of 120 hours. The company manufacturing the bulbs claims that the average life of their bulbs is 1600 hrs. Is the claim acceptable at 5% level of significance?	**06**	**L3**	**CO4**
	b.	The two independent samples of eight and seven items respectively had the following values of the variable: Sample 1: 9, 11, 13, 11, 15, 9, 12, 14 Sample 2: 10, 12, 10, 14, 9, 8, 10 Do the two estimates of population variance differ significantly at 5% level of significance? F at 5% $(V_1 = 7, V_2 = 6) = 4.21$.	**07**	**L3**	**CO4**
	c.	Table gives the number of aircraft accidents that occurred during the various days of a week. Test whether the accidents are uniformly distributed over the week. $\chi^2_{5\%}(\gamma = 5) = 11.07$. Day: Mon, Tue, Wed, Thur, Fri, Sat Number of accidents: 15, 19, 13, 12, 16, 15	**07**	**L3**	**CO4**

OR

Q.8 **a.** Two random samples gave the following data: — **06** | **L2** | **CO4**

	Size	Mean	Variance
Sample 1	8	9.6	1.2
Sample 2	11	16.5	2.5

Can we conclude that the two samples have been drawn from the same normal population? $F_{5\%}$ (10, 7) = 3.64.

b. The following data relate to the marks obtained by 11 students in two tests. Second test is after intense coaching. Do the data indicate that the students have benefited by coaching? — **07** | **L3** | **CO4**

Test 1	19	23	16	24	17	18	20	18	21	19	20
Test 2	17	24	20	24	20	22	20	20	18	22	19

($t_{5\%}$ (γ = 10) is 1.81)

c. The mean value of a random sample of 60 items was found to be 145 and standard deviation is 40. Find the 95% confidence limits for the population mean. — **07** | **L2** | **CO5**

Module – 5

Q.9 **a.** The following figures relate to production in kgs of three variables A, B, C of wheat sown on 12 plots. — **10** | **L3** | **CO6**

A	14	16	18		
B	14	13	15	22	
C	18	16	19	19	22

Apply one-way Annova using a 0.05 significance level in the production of the varieties. F_c at 5% (2, 9) d.f is 4.26.

b. Analyze and interpret the following statistics concerning output of wheat per field obtained as a result of experiment conducted to test four varieties of wheat viz., A, B, C and D under a Latin - Square design. — **10** | **L3** | **CO6**

C	B	A	D
25	23	20	20
A	D	C	B
19	19	21	18
B	A	D	C
19	14	17	20
D	C	B	A
17	20	21	15

OR

Q.10 **a.** Four doctors each test four treatments for a certain disease and observe the number of days each patient takes to recover. The results are as follows: — **10** | **L3** | **CO6**

Doctor/Treatment	T_1	T_2	T_3	T_4
D_1	10	14	19	20
D_2	11	15	17	21
D_3	9	12	16	19
D_4	8	13	17	20

Discuss the difference between doctors and treatments F_{at} 5% level (3, 9) is 3.86.

	b.	A study of the effect of different types of anesthesia on the length of post-operative hospital stay yielded for the following for cesarean patients. Group 'A' was given an epidural MS providing additional safety. Group 'B' was given an epidural and Group 'C' was given a spinal is considered to be less dangerous and Group 'D' was given general anesthesia is considered to be the most dangerous. Note that the data are in the form of distribution for each group.	**10**	**L3**	**CO6**

	Length of Stay	Number of patients
Group A	3	6
	4	14
Group B	4	18
	5	2
Group C	4	10
	5	9
	6	1
Group D	4	8
	5	12

Test for the existence of an effect due to anesthesia type at 0.01. $F_{0.01}$ = 4.13

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